Neo taxonomical tools – This is based on Electron Microscopy images to study the molecular structures of cell organelles.

Ethology of taxonomical tools – Based on the behaviour of the organisms it can be classified. For example sound of birds, bioluminescence, etc.

e-Taxonomic resources – INOTAXA is an electronic resource for digital images and description about the species which was developed by Natural History Museum, London. **INOTAXA** means **In**tegrated **O**pen **TAX**onomic **A**ccess.

Summary

Earth has numerous habitats with a wide range of living organisms inhabiting it. Living organisms show a variety of unique characters different from non-living matter. Classification is the process by which anything is grouped in a convenient category based on some easily observable characters.

Taxonomy is the science of arrangement of living organisms. R. H. Whittaker proposed the five kingdom classification. Three domain classification was proposed by Carl Woese and his co-workers.

The taxonomical hierarchy includes seven categories namely kingdom, phylum, class, order, family, genus and species. The process of assigning scientific names to animal or taxonomic group is called nomenclature. Each scientific name has two components, generic name and a specific epithet. The important component of the taxonomical tools are field visits, survey, identification, classification, preservation and documentation. Molecular taxonomical tools are more accurate, authentic and significant for taxonimical classification.

Activity

The main objective of this activity is to check the students understanding about animals and its characteristics before learning the lesson. Observe the picture given below, identify the animals and classify them according to your own understanding; write one character about each class of animals.



Take the students to the school ground and ask them to observe and identify few invertebrates (insects, earthworm, spiders etc). Ask the students to write few characteristics of each animal which they have observed.

Sl.No.	Name of the Animal	Known Character	Class	Habitat
1				
2				
3				
4				

XI Std Zoology Chapter-1 EM.indd 12 30/12/2022 10:28:50